

Benjamin Aaron Storer

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Education

2013–2018 Ph. D., Applied Mathematics, University of Waterloo
Thesis: *Development and Application of Models and Diagnostics for Geophysical Fluid Flows*
Supervisor: Francis Poulin
2008–2013 B. Math, Honours Pure Mathematics with distinction
Co-operative Program, Applied Mathematics Minor
University of Waterloo

Teaching

2019-2026	ME201, Applied Boundary Value Problems	Univ. of Rochester	Co-instructor
2016	MATH127, Calculus 1 for the Sciences	Univ. of Waterloo	Instructor
2011	TMAT100, Technical Mathematics	Humber College	Instructor

Academic Positions

Jan 2026 - Present	Research Assistant Professor	University of Rochester Mentor: Hussein Aluie
Feb 2023 - Dec 2025	Research Associate	University of Rochester Mentor: Hussein Aluie
Feb 2019 - Jan 2023	Postdoctoral Research Associate	University of Rochester Mentor: Hussein Aluie

Software Development

Grants / Funding

[1] **ACCESS Maximize Computing Grant (EES220052)** Role: P.I.

- 6-month preliminary, awarded 8 million core hours. Estimated value of \$33,280.00 USD.
- 12-month renewal, awarded 58 million core hours. Estimated value of \$268,885.00 USD.

FlowSieve

Role : primary developer

Description: Highly parallelized HPC codebase for analysing multi-scale energetics of oceanic flows, and is broadly applicable to flows in other spherical settings (atmospheric, stellar, etc). Works in spherical geometries and has built-in diagnostics for extracting metrics of energy and enstrophy cascades. Includes Helmholtz decomposition tools for analysing general flow fields.

Links: [Documentation](#), [Github Repository](#), [JOSS Article](#)

Invited Presentations

Invited Keynote Speaker 2024 DRAKKAR Ocean Modelling Workshop

FlowSieve: Coarse-Graining to Gain Insight into Multi-Scale Dynamics

29-31 January 2024 Grenoble, France

Invited Speaker Department of Physics, University of Rome

Studying the Multi-Scale Dynamics of the Oceanic Symphony

18 April 2023, Rome, Italy

Invited Speaker Italian National Research Council (CNR)

Institute of Marine Sciences (ISMAR)

Studying the Multi-Scale Dynamics of the Oceanic Symphony

20 April 2023, Rome, Italy

Conference Organization

Session Organizer “*AI006 - Oceanic and Atmospheric Interactions and Drivers: Their Scale-Dependence and Climate Impacts*”

Chairs : **Storer, B.**, O'Neill, L, Buzzicotti, M, Kido, S., and Hayden, E.

Biennial Ocean Sciences Meeting, New Orleans, Louisiana, USA, February 2024.

Tutorial Session “*Coarse-graining: blurring complex oceanic flows for insight*”

Chairs : Aluie, H., **Storer, B.**

Biennial Ocean Sciences Meeting, Honolulu, Hawaii, February 2022.

Community Engagement and Outreach

Flower City Pride Band An LGBTQ+ community band based in Rochester, NY
501(c)(3) registered charitable organization
Role: Assistant Artistic Director (2022 - present)
Board of Directors (Secretary) (2022 - present)
Member (2019 - present)

Publications

- [1] Hadi Sichani, P., **Storer, B. A.**, & Aluie, H. (2026). *Atmospheric transfer of kinetic energy across planetary-scale circulation and weather*. Physical Review Research, 8(3), 033239.
- [2] Kouhen, S., **Storer, B. A.**, Aluie, H., Marhsall, D. P., & Christensen, H. M. (2024) *Convective and orographic origins of the mesoscale kinetic energy spectrum*. Geophysical Research Letters, 51(21).
- [3] Xue, S., **Storer, B. A.**, Glade, R. C., & Aluie (2024). *Surface Variability Mapping and Roughness Analysis of the Moon Using a Coarse-Graining Decomposition*. Journal of Geophysical Research: Planets, 129(10).
- [4] Khatri, H., Griffies, S. M., **Storer, B. A.**, Buzzicotti, M., Aluie, H., Sonnewald, M., Dussin, R., & Shao, A. E. (2024). *A scale-dependent analysis of the barotropic vorticity budget in a global ocean simulation*. Journal of Advances in Modeling Earth Systems, 16, 1–29.
- [5] **Storer, B. A.**, Buzzicotti, M., Khatri, H., Griffies, S. M., & Aluie, H. (2023). *Global cascade of kinetic energy in the ocean and the atmospheric imprint*. Science Advances, 9(51).
- [6] Buzzicotti, M., **Storer, B. A.**, Khatri, H., Griffies, S. M., & Aluie, H. (2023). *Spatio-Temporal Coarse-Graining Decomposition of the Global Ocean Geostrophic Kinetic Energy*. Journal of Advances in Modeling Earth Systems, 15(6).
- [7] **Storer, B. A.** & Aluie, H. (2023). *FlowSieve: A Coarse-Graining Utility for Geophysical Flows on the Sphere*. Journal of Open Source Software, 8(84), 4277
- [8] **Storer, B. A.**, Buzzicotti, M., Khatri, H., Griffies, S. M., & Aluie, H. (2022). *Global energy spectrum of the general oceanic circulation*. Nature Communications, 13(1), 5314.
- [9] Magill M., Coutino A., **Storer, B. A.**, Stastna, M., & Poulin, F. J. (2019). *Dynamics of nonlinear Alfvén waves in the shallow water magnetohydrodynamic equations*. Physical Review Fluids, 4(5), 053701.
- [10] **Storer, B. A.**, Poulin, F. J., & Ménesguen, C. (2018). *The Dynamics of Quasigeostrophic Lens-Shaped Vortices*. Journal of Physical Oceanography, 48, 937–957.
- [11] Poulin, F. J., Borrisov, A., **Storer, B. A.**, & Stastna, M. (2018). *A shallow water model of the solar tachocline: A numerical approach to determine wave structure*. Dynamics of Continuous, Discrete and Impulsive Systems Series B: Applications and Algorithms, 25(3–4), 219–231.
- [12] Willick, K., **Storer, B. A.**, & Wesolkowski, S. (2013). *A new principal curve algorithm and standard deviation clouds for non-parametric ordered data analysis*. In 2013 IEEE Congress on Evolutionary Computation, CEC 2013 (pp. 1459–1466).